Gautret et al. 2020 (retracted): Exercises 1 and 2

Workshop handout: for your own use, do not post online. Page numbers = pages of the original document.

- error to catch
- ask the authors / weakness
- good practice / not an error
- reference passage (registry, SAP, methods)

C7 (p. 1) Viral LOAD is promised but the outcome is binary PCR positivity; no quantitative load reported.

O7 (p. 1) The abstract itself says single-arm, yet the paper reports a controlled comparison.

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O6 (p. 1) Register says no control group. Controls are refusers and patients from other centres: not a comparable group.

O4 (p. 1) Day 6 is not a registered time point (register: days 1, 4, 7, 14). End point chosen after data were seen?

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F3 (p. 3) A death is excluded from an efficacy analysis of the treated arm.

F4 (p. 3) Asymmetric: losses only in the treated arm.

C4 (p. 3) 14-year age gap, p = 0.06 read as 'no difference'. Baseline p-values do not test confounding; viral load at baseline not reported.

O5 (p. 3) Primary end point = day 6; register lists days 1/4/7/14; the sample size below uses day 7.

S1 (p. 3) Powered on DAY 7 and 48 patients (24 per group). Reported: day 6, 36 analysed (20 vs 16). Neither matches the register (25).

M1 (p. 3) Four days x several comparisons, no correction, n = 36. Any single p loses its meaning.

F1 (p. 3) 42 enrolled -> 26 HCQ + 16 controls -> 6 HCQ removed -> 20 vs 16. No flow diagram.

F2 (p. 3) Not 'lost': 3 ICU, 1 death, 1 discharge, 1 nausea. All in the treated arm, all bad outcomes, all removed. Outcome-dependent exclusion (ROBINS-I missing data: Critical).

M3 (p. 3) 6/6: a percentage on 6 non-randomised patients.

C5 (p. 4) Outcome measured differently by arm; missing days in controls imputed as POSITIVE. Biases towards a treatment effect (ROBINS-I measurement).

M2 (p. 4) Subgroup claim with data not shown: cannot be checked.

S3 (p. 5) 'So significant and evident': unplanned early release justified by the result itself.

P2 (p. 5) Causal efficacy claim; design cannot support it.

P4 (p. 5) Treatment recommendation from an interim, uncontrolled, surrogate-outcome analysis. Contrast with the honest rewrite in keys.pdf.

F5 (p. 5) Limitations call them 'dropouts', not deaths and ICU transfers.

P5 (p. 5) In vitro data do not corroborate a confounded clinical comparison.

O2 (p. 6) 'None declared' although the editor-in-chief is an author and the institute promoted the drug.

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Hydroxychloroquine and azithromycin as a treatment of COVID-19: results of an open-label non-randomized clinical trial ☆

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ABSTRACT

Backgr[ound:] [C]hloroqu[ine] and hydroxychloroquine have been found to be efficient on SARS-CoV-2, and rep[orted] to [be] efficient [in] Chinese COV-19 patients. We evaluate the effect of hydroxychloroquine on [respi]ratory v[iral] loads.

Patients and methods: French Confirmed COVID-19 patients were included in a single arm protocol fro[m] March to [M]arch 16th, to receive 600mg of hydroxychloroquine daily and their viral load in nasop[ha]ryngeal swabs was tested daily in a hospital setting. Depending on their clinical presentation, azithromycin [wa]s added to the treatment. Untreated patients from another center and cases refusing the protocol we[re] included as negative controls. Presence and absence of virus at Day6-post inclusion was considered the end point.

Results: Six patients were asymptomatic, 22 had upper respiratory tract infection symptoms and eight ha[d] lower respiratory tract infection symptoms.

Twenty cases were treated in this study and showed a significant reduction of the viral carriage at D6-post inclusion compared to controls, and much lower average carrying duration than reported in the litterature for untreated patients. Azithromycin added to hydroxychloroquine was significantly more efficient for virus elimination.

Conclusion: Despite its small sample size, our survey shows that hydroxychloroquine treatment is significantly associated with viral load reduction/disappearance in COVID-19 patients and its effect is reinforced by azithromycin.

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1. Introduction

In late December 2019, an outbreak of an emerging disease (COVID-19) due to a novel coronavirus (later named SARS-CoV-2) started in Wuhan, China and rapidly spread in China and outside [1,2]. The WHO declared the epidemic of COVID-19 as a pandemic on March 12th 2020 [3]. According to a recent Chinese study,

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☆ Given his role as Editor in Chief of this journal, Jean Marc Rolain had no involvement in the peer-review of this article and has no access to information regarding its peer-review. Full responsibility for the peer-review process for this article was delegated to P.R. Hsueh.

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$ equal work

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about 80% of patients present with mild disease and the overall case-fatality rate is about 2.3% but reaches 8.0% in patients aged 70 to 79 years and 14.8% in those aged ≥80 years [4]. However, there is probably an important number of asymptomatic carriers in the population, and thus the mortality rate is probably overestimated. France is now facing the COVID-19 wave with more than 4500 cases, as of March 14th 2020 [5]. Thus, there is an urgent need for an effective treatment to treat symptomatic patients but also to decrease the duration of virus carriage in order to limit the transmission in the community. Among candidate drugs to treat COVID-19, repositioning of old drugs for use as antiviral treatment is an interesting strategy because knowledge on safety profile, side effects, posology and drug interactions are well known [6,7].

A recent paper reported an inhibitor effect of remdesivir (a new antiviral drug) and chloroquine (an old antimalarial drug) on the growth of SARS-CoV-2 *in vitro,* [8] and an early clinical trial conducted in Chinese COVID-19 patients, showed that chloroquine had a significant effect, both in terms of clinical outcome and viral clearance, when compared to controls groups [9,10]. Chinese experts recommend that patients diagnosed with mild, moderate and severe cases of COVID-19 pneumonia and without contraindications to chloroquine, be treated with 500 mg chloroquine twice a day for ten days [11].

Hydroxychloroquine (an analogue of chloroquine) has been demonstrated to have an anti-SARS-CoV activity *in vitro* [12]. Hydroxychloroquine clinical safety profile is better than that of chloroquine (during long-term use) and allows a higher daily dose [13] and has fewer concerns regarding drug-drug interactions [14]. Our team has a very comprehensive experience (over 20 years) in successfully treating patients with chronic diseases due to intracellular bacteria (Q fever due to *Coxiella burnetii* and Whipple's disease due to *Tropheryma whipplei*) with long-term hydroxychloroquine treatment (600 mg/day for 12 to 18 months) [15,16]. We therefore started to conduct a clinical trial aiming at assessing the effect of hydroxychloroquine on SARS-CoV-2-infected patients after approval by the French Ministry of Health. In this report we describe our early results, focusing on virologic[al] [e]ffect of hydroxychloroquine receiving hydroxychloroquine as compared to a control group.

2. Study population and Methods

2.1. Setting

This ongoing study is coordinated by the Méditerranée Infection University Hospital Institute in Marseille. Patients who were proposed a treatment with hydroxychloroquine were recruited and managed in the Marseille centre. Controls without hydroxychloroquine treatment were recruited in Marseille, Nice, Avignon and Briançon centers located in Southern France.

2.2. Patients

Hospitalized patients with confirmed COVID-19 were included in this study if they fulfilled two primary criteria: i) age >12 years; ii) PCR documented SARS-CoV-2 carriage in nasopharyngeal sample at admission regardless of their clinical status.

Patients were excluded if they had a known allergy to hydroxychloroquine or chloroquine or had another known contraindication to treatment with the study drug, including retinopathy, G6PD deficiency and QT prolongation. Breastfeeding and pregnant patients were excluded based on their declaration and pregnancy test results when required.

2.3. Informed consent

Before being included in the study, patients meeting inclusion criteria had to give their consent to participate to the study. Written informed signed consent was obtained from adult participants (≥ 18 years) or from parents or legal guardians for minors (<18 years). An information document that clearly indicates the risks and benefits associated with the participation to the study was given to each patient. Patients received information about their clinical status during care regardless of whether they participate in the study or not. Regarding patient identification, a study number was assigned sequentially to included participants, according to the range of patient numbers allocated to the study centre. The study was conducted in accordance with the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) guidelines on good clinical practice, the Helsinki Declaration, and applicable standard operating procedures.

The protocol, appendices and any other relevant documentation were submitted to the French National Agency for Drug Safety (ANSM) (2020-000890-25) and to the French Ethic Committee (CPP Ile de France) (20.03.[illegible].99113), reviewed and were approved on 5th and 6th March 2020, respectively. This trial is registered in EU Clinical Trials Register with number 2020-000890-25.

2.4. Procedures

Patients were seen at baseline for enrollment, initial data collection and treatment at day-0, and again for daily follow-up during a 14 day period. Each day, patients received a standardized clinical examination and when possible, a nasopharyngeal sample was collected. All clinical data were collected using standardized questionnaires. All patients in the Marseille centre were proposed oral hydroxychloroquine sulfate 200 mg, three times per day for ten days (in this preliminary phase, we did not enroll children in the treatment group based on data indicating that children develop mild symptoms of COVID-19 [4]). Patients who refused the treatment or had an exclusion criteria, served as controls in the Marseille centre. Patients in other centers did not receive hydroxychloroquine and served as controls. Symptomatic treatment and antibiotics as a measure to prevent bacterial super-infection was provided by investigators based on clinical judgment. Hydroxychloroquine was provided by the National Pharmacy of France on nominative demand.

2.5. Clinical classification

Patients were grouped into three categories: asymptomatic, upper respiratory tract infection (URTI) when presenting with rhinitis, pharyngitis, or isolated low-grade fever and myalgia, and lower respiratory tract infections (LRTI) when presenting with symptoms of pneumonia or bronchitis.

2.6. PCR assay

SARS-CoV-2 RNA was assessed by real-time reverse transcription-PCR [17].

2.7. Hydroxychloroquine dosage

Native hydroxychloroquine has been dosed from patients' serum samples by UHPLC-UV using a previously described protocol [18]. The peak of the chromatogram at a retention time of 1.05 min corresponds to hydroxychloroquine metabolite. The serum concentration of this metabolite is deduced from UV absorption, as

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S2 — Unplanned interim analysis, no stopping rule; trial 'ongoing'.

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C3 — Children only in the control arm (age > 12 eligible): age differs by design.

C2 — Controls include refusers and patients with contraindications: selection into treatment.

Table 1
Characteristics of the study population.

	Age (years)			Male gender		Clinical status				Time between onset of symptoms and inclusion (days)		
	Mean ± SD	t	p-value	n (%)	p-value	Asymptomatic	URTI	LRTI	p-value	Mean ± SD	t	p-value
Hydroxychloroquine treated patients (N=20)	51.2 ± 18.7	-1.95	0.06	9 (45.0)	0.65	2 (10.0)	12 (60.0)	6 (30.0)	0.30	4.1 ± 2.6	-0.15	0.88
Control patients (N=16)	37.3 ± 24.0			6 (37.5)		4 (25.0)	10 (62.5)	2 (12.5)		3.9 ± 2.8		
All patients (36)	45.1 ± 22.0			15 (41.7)		6 (16.7)	22 (61.1)	8 (22.2)		4.0 ± 2.6		

URTI: upper tract respiratory infection, LRTI: lower tract respiratory infection

for hydroxychloroquine concentration. Considering both concentrations provides an estimate of initial serum hydroxychloroquine concentration.

2.8. Culture

For all patients, 500 µL of the liquid collected from the nasopharyngeal swab were passed through a 0.22-µm pore sized centrifugal filter (Merck millipore, Darmstadt, Germany), then were inoculated in wells of 96-well culture microplates, of which 4 wells contained Vero E6 cells (ATCC CRL-1586) in Minimum Essential Medium culture medium with 4% fetal calf serum and 1% glutamine. After centrifugation at 4,000 g, the microplates were incubated at 37°C. Plates were observed daily for evidence of cytopathogenic effect. Presumptive detection of the virus in the supernatant was done using SU5000 SEM (Hitachi) then confirmed by specific RT-PCR.

2.9. Outcome

The primary endpoint was virological clearance at day-6 post-inclusion. Secondary outcomes were virological clearance at other time of the study period and clinical follow-up (fever, temperature, respiratory rate, lenght of stay in hospital and mortality), and occurrence of side-effects.

2.10. Statistics

Assuming a 50% efficacy of hydroxychloroquine in reducing the viral load at day 7, an alpha risk, a type I error rate of 5% and 10% loss to follow-up, we calculated that a total of 48 COVID-19 patients (ie, 24 in the hydroxychloroquine group and 24 in the control group) would be required for the analysis (Fleiss with CC). Statistical differences were evaluated by Pearson's chi-square or Fisher's exact test for categorical variables, as appropriate. Means of quantitative data were compared using Student's t-test. Analyses were performed using Stata version 14.2.

3. Results (detailed results are available in supplementary Table 1)

3.1. Demographics and clinical presentation

We enrolled 36 out of 42 patients meeting the inclusion criteria in this study that had at least six days of follow-up at the time of the present analysis. A total of 26 patients received hydroxychloroquine and 16 were control patients. Six hydroxychloroquine-treated patients were lost during the follow-up of the survey because of early cessation of treatment. Reasons are as follows: three patients were transferred to intensive care unit, including one transferred on day2 post-inclusion who was PCR-positive on day1, one transferred on day3 post-inclusion who was PCR-positive on days1-2 and one transferred on day4 post-inclusion who was PCR-positive on day1 to day3, one patient died on day3 post inclusion and was PCR-positive on day3, one patient decided to leave the hospital on day3 post-inclusion and was PCR-negative on days1-2; finally, one patient stopped the treatment on day3 post-inclusion because of nausea and was PCR-positive on days1-2-3. The results presented here are therefore those of 36 patients (20 hydroxychloroquine-treated patients and 16 control patients). No one of the control patients were lost during follow-up. Basic demographics and clinical status are presented in Table 1. Overall, 15 patients were male (41.7%), with a mean age of 45.1 years. The proportion of asymptomatic patients was 16.7%, that of patients with URTI symptoms was 61.1% and that of patients with LRTI symptoms was 22.2%. All patients with LRTI symptoms had confirmed pneumonia by CTScan. Hydroxychloroquine-treated patients were older than control patients (51.2 years vs. 37.3 years). No significant difference was observed between hydroxychloroquine-treated patients and control patients with regard to gender, clinical status and duration of symptoms prior to inclusion (Table 1). Among hydroxychloroquine-treated patients six patients received azithromycin (500mg on day1 followed by 250mg per day for the next four days) to prevent bacterial super-infection and were kept under daily electrocardiogram control. Clinical follow-up and occurrence of side-effects will be described in a further paper at the end of the trial.

3.2. Hydroxychloroquine dosage

Mean hydroxychloroquine serum concentration was 0.46 µg/ml±0.2 (N=20).

3.3. Effect of hydroxychloroquine on viral load

The proportion of patients that had negative PCR results in nasopharyngeal samples significantly differed between treated patients and controls at days 3-4-5 and 6 post-inclusion (Table 2). At day6 post-inclusion, 70% of hydroxychloroquine-treated patients were virologically cured compared to 12.5% in the control group (p= 0.001).

When comparing the effect of hydroxychloroquine treatment as a single drug and the effect of hydroxychloroquine and azithromycin in combination, the proportion of patients that had negative PCR results in nasopharyngeal samples was significantly different between the two groups at days 3-4-5 and 6 post-inclusion (Table 3). At day6 post-inclusion, 100% of patients treated with a combination of hydroxychloroquine and azithromycin were virologically cured compared to 57.1% of patients treated with hydroxychloroquine only, and 12.5% in the control group (p<0.001).

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F4 — Asymmetric: losses only in the treated arm.

C4 — 14-year age gap, p = 0.06 read as 'no difference'. Baseline p-values do not test confounding; viral load at baseline not reported.

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S1 — Powered on DAY 7 and 48 patients (24 per group). Reported: day 6, 36 analysed (20 vs 16). Neither matches the register (25).

F1 — 42 enrolled -> 26 HCQ + 16 controls -> 6 HCQ removed -> 20 vs 16. No flow diagram.

F2 — Not 'lost': 3 ICU, 1 death, 1 discharge, 1 nausea. All in the treated arm, all bad outcomes, all removed. Outcome-dependent exclusion (ROBINS-I missing data: Critical).

M1 — Four days x several comparisons, no correction, n = 36. Any single p loses its meaning.

M3 — 6/6: a percentage on 6 non-randomised patients.

Table 2
Proportion of patients with virological cure (negative nasopharyngeal PCR) by day, in COVID-19 patients treated with hydroxychloroquine and in COVID-19 control patients.

	Day3 post inclusion			Day4 post inclusion			Day5 post inclusion			Day6 post inclusion		
	Number of negative patients/total number of patients	%	p-value	Number of negative patients/total number of patients	%	p-value	Number of negative patients/total number of patients	%	p-value	Number of negative patients/total number of patients	%	p-value
Hydroxychloroquine treated patients (N=20)	10/20	50.0	0.005	12/20	60.0	0.04	13/20	65.0	0.006	14/20	70.0	0.001
Control patients (N=16)	1/16	6.3		4/16	25.0		3/16	18.8		2/16	12.5	

[a]control patients from centers other than Marseille did not undergo daily sampling, but were sampled every other day in most cases, t[illegible]were c[illegible]red positive for PCR when actually positive the day(s) before and the day(s) after the day(s) with missing data.

Outcome measured differently by arm; missing days in controls imputed as POSITIVE. Biases towards a treatment effect (ROBINS-I measurement).

C5

Table 3
Proportion of patients with virological cure (negative nasopharyngeal PCR) by day, in COVID-19 patients treated with hydrox[illegible]roquine only, [illegible]VID[illegible]patients treated with hydroxychloroquine and azithromycin combination, and in COVID-19 control patients.

	Day3 post inclusion			Day4 post inclusion			Day5 post inclusion			Day6 post inclusion		
	Number of negative patients/total number of patients	%	p-value	Number of negative patients/total number of patients	%	p-value	Number of negative patient[illegible] num[illegible]of patients	%	[illegible]e	Nu[illegible]of p[illegible]ve [illegible]ients/total [illegible]number of patients	%	p-value
Control patients	1/16	6.3	0.002	4/16	25.0	0.05		18.[illegible]	[illegible]002	2/16	12.5	<0.001
Hydroxychloroquine treatment only	5/14	35.7		7/14	50.0		7/14	50.0		8/14	57.1	
Hydroxychloroquine and azithromycin combined treatment	5/6	83.3		5/6	83.3		6/6	100		6/6	100	

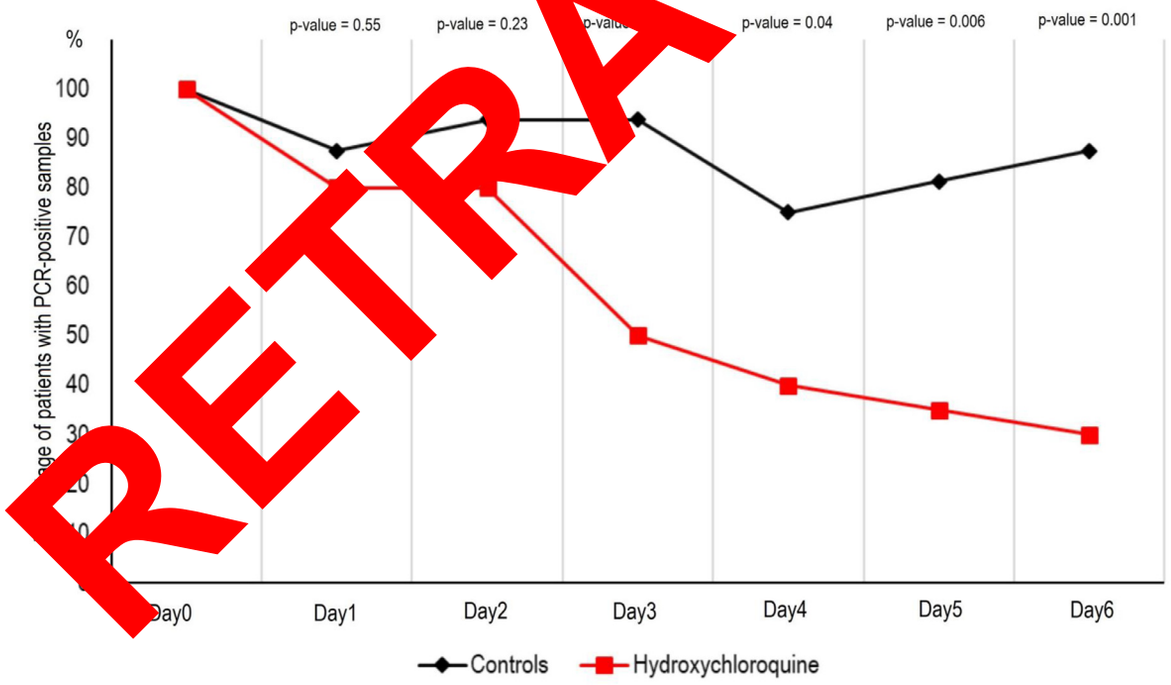


Fig. 1. Percentage of patients with PCR-positive nasopharyngeal samples from inclusion to day6 post-inclusion in COVID-19 patients treated with hydroxychloroquine and in COVID-19 control patients.

Subgroup claim with data not shown: cannot be checked.

M2

These results are summarized in Figs. 1 and 2. Drug effect was significantly higher in patients with symptoms of URTI and LRTI, as compared to asymptomatic patients with $p<0.05$ (data not shown).

Of note, one patient who was still PCR-positive at day6-post inclusion under hydroxychloroquine treatment only, received azithromycin in addition to hydroxychloroquine at day8-post inclusion and cured her infection at day-9 post inclusion. In contrast, one of the patients under hydroxychloroquine and azithromycin combination who tested negative at day6 post-inclusion was tested positive at low titer at day8 post-inclusion.

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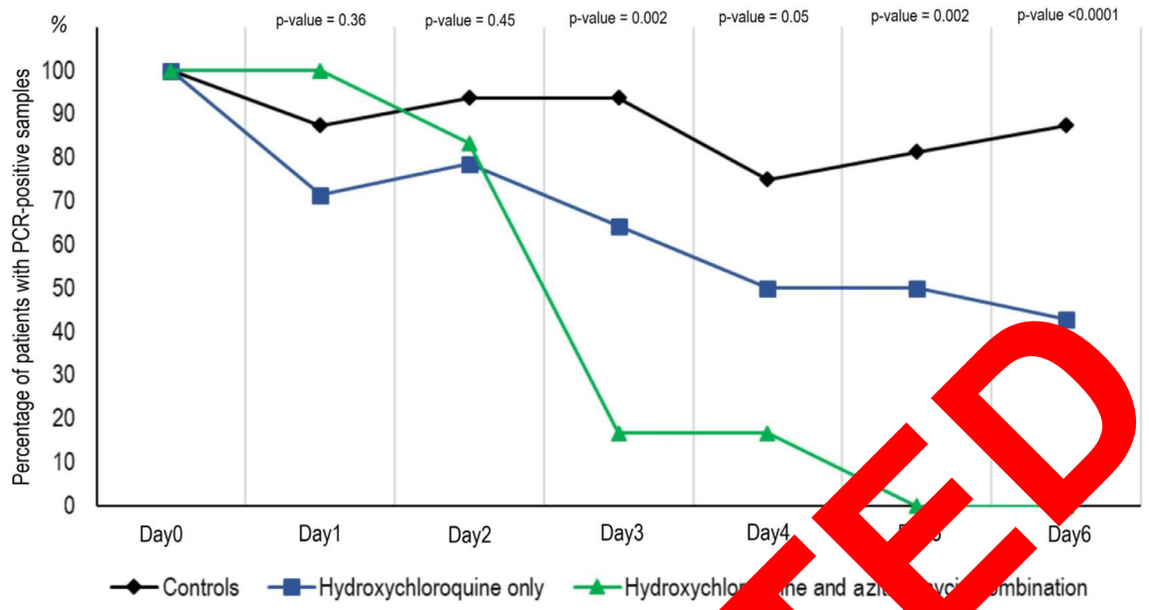


Fig. 2. Percentage of patients with PCR-positive nasopharyngeal samples from inclusion to day6 pos[t]-inclusion in COVID-19 patients treated with hydroxychloroquine only, in COVID-19 patients treated with hydroxychloroquine and azithromycin combination, and in COVID-19 control pa[tients].

3.4. Cultures

We could isolate SARS-CoV-2 in 19 out of 25 clinical samples from patients.

4. Discussion

For ethical reasons and because our first results are so sig[nifi]cant and evident we decided to share our findin[gs] with the medical community, given the urgent need for an e[ffe]ctive c[ur]e against SARS-CoV-2 in the current pandemic conte[xt].

We show here that hydroxychloroqu[ine] is highly effic[acious] in reducing viral nasopharyngeal carriage of SARS-CoV-2 in [COVID]-19 patients in only three to six days, in mos[t] patients. A sig[nifi]cant difference was observed between hydroxychloroquine-trea[ted] patients and controls starting even from day3 post-inclusion. These results are of great importance be[cau]se a recent p[aper] has shown that the mean duration of viral shedding in patients s[uffe]ring from COVID-19 in China was 20 days (up t[o] 37 days for the longest duration) [19]

Very recently, a Chi[nese] team published results of a study demonstrating in vitro that chloroquine and hydroxychloroquine inhibit SARS-CoV-2 *in vitro* with hydroxychloroquine (EC50=0.72%µM) found to be more potent than chloroquine (EC50=5.47%µM) [14]. These *in vitro* results corroborate our clinical results. The target values indicated in this paper [14] were reached in our experiments. The safe dose-dependent toxicity profile of hydroxychloroquine in humans, compared to that of chloroquine [13], allows using clinical doses of hydroxychloroquine that will be over its EC50 observed *in vitro* [14].

Our preliminary results also suggest a synergistic effect of the combination of hydroxychloroquine and azithromycin. Azithromycin has been shown to be active *in vitro* against Zika and Ebola viruses [20-22] and to prevent severe respiratory tract infections when administrated to patients suffering from viral infection [23]. This finding should be further explored to know whether a combination is more effective especially in severe cases. Speculated potential risk of severe QT prolongation induced by the association of the two drugs has not been established yet but should be considered. As for each treatment, the cost benefits of the risk should be evaluated individually. Further studies on this combination are needed, since such combination may both act as an antiviral therapy against SARS-CoV-2 and prevent bacterial super-infections.

If cause[d] by a failure of hydroxychloroquine treatment should be investigated by testing the isolated SARS-CoV-2 strains of the non-respondents and analyzing their genome, and by analyzing the factors that may be associated with the metabolism of hydroxychloroquine. The existence of hydroxychloroquine failure in two patients (mother and son) is more suggestive of the last mechanism of resistance.

Such results are promising and open the possibility of an international strategy to decision-makers to fight this emerging viral infection in real-time even if other strategies and research including vaccine development could be also effective, but only in the future. We therefore recommend that COVID-19 patients be treated with hydroxychloroquine and azithromycin to cure their infection and to limit the transmission of the virus to other people in order to curb the spread of COVID-19 in the world. Further works are also warranted to determine if these compounds could be useful as chemoprophylaxis to prevent the transmission of the virus, especially for healthcare workers. Our study has some limitations including a small sample size, limited long-term outcome follow-up, and dropout of six patients from the study, however in the current context, we believe that our results should be shared with the scientific community.

Acknowledgements

We thank Céline Boschi, Stéphanie Branger, Véronique Filosa, Géraldine Gonfrier, Nadège Palmero, Magali Richez and all the clinical, technical and paramedical staff of the hospitalization units and laboratories for their support in this difficult context.

Declarations

Funding: This work was supported by the French Government under the « Investissements d'avenir » (Investments for the Future) program managed by the Agence Nationale de la Recherche (ANR,

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fr: National Agency for Research), (reference: Méditerranée Infection 10-IAHU-03).

'None declared' although the editor-in-chief is an author and the institute promoted the drug.

02

Competing Interests: None declared.

Ethical Approval: French Ethic Committee (CPP Ile de France) (20.02.28.99113).

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijantimicag.2020.105949.

References

[1] Lai CC, Shih TP, Ko WC, Tang HJ, Hsueh PR. Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and coronavirus disease-2019 (COVID-19): The epidemic and the challenges. Int J Antimicrob Agents 2020 Feb 17:105924 [Epub ahead of print]. doi:10.1016/j.ijantimicag.2020.105924.

[2] Wang LS, Wang YR, Ye DW, Liu QQ. A review of the 2019 Novel Coronavirus (COVID-19) based on current evidence". Int J Antimicrob Agents 2020 [Epub ahead of print].

[3] WHO Director-General's opening remarks at the media briefing on COVID-19 - 11 March 2020. [https://www.who.int/dg/speeches/detail/who-director-general-s-opening-remarks-at-the-media-briefing-on-covid-19---11-march-2020]

[4] Wu Z, McGoogan JM. Characteristics of and important lessons from the coronavirus disease 2019 (COVID-19) outbreak in China: summary of a report of 72 314 cases from the Chinese Center for Disease Control and Prevention. JAMA 2020 Feb 24 [Epub ahead of print]. doi:10.1001/jama.2020.2648.

[5] Santé Publique France. Infection au nouveau Coronavirus (SARS-CoV-2), COVID-19, France et Monde [https://www.santepubliquefrance.fr/maladies-et-traumatismes/maladies-et-infections-respiratoires/infection-a-coronavirus/articles/infection-au-nouveau-coronavirus-sars-cov-2-covid-19-france-et-monde]

[6] Colson P, Rolain JM, Raoult D. Chloroquine for the 2019 novel coron SARS-CoV-2. Int J Antimicrob Agents 2020 Feb 15:105923 [Epub ahea print]. doi:10.1016/j.ijantimicag.2020.105923.

[7] Colson P, Rolain JM, Lagier JC, Brouqui P, Raoult D. Chloroquine and hydroxychloroquine as available weapons to fight COVID-19. Int J Antimicrob Agent 2020 [Epub ahead of print].

[8] Wang M, Cao R, Zhang L, Yang X, Liu J, Xu M, et al. R sivir chloroquine effectively inhibit the recently emerged novel avirus (-nCoV) in vitro. Cell Res 2020:10–0282.

[9] Gao J, Tian Z, Yang X. Breakthrough: Chloroquine phos ha ent efficacy in treatment of COVID-19 associat eumo inical studies. Biosci Trends 2020 Feb 19 [Epub ahead of p , doi:10.58 2020.01047.

[10] Chinese Clinical Trial Registry. htt chictr.org.cn/ proj.aspx?title=%E6%B0%AF%E5%96%B9&officialna id=&secondary &applier=&studyleader=&ethicalcomm esanc ponsor=&studyailment=&studyailmentcode=&studytype= tudystage=0& esign=0&minstudyexecutetime=&ma yexecutetime=&re entstatus=0&gender=0&agreetosign= onsor=&regno=&regstat &country=&province=&city=&institu &institu evel=&measure=&intercode=&sourceofspends=&crea =0&is urf=&whetherpublic=&btngo=btn&verifycode=&page=1.

[11] Multicenter collaboration group of Department of Science and Technology of Guangdong Province and Health Commission of Guangdong Province for chloroquine in the treatment of novel coronavirus pneumonia Expert consensus on chloroquine phosphate for the treatment of novel coronavirus pneumonia]. Zhonghua Jie He He Hu Xi Za Zhi 2020 Mar 12;43(3):185–8. doi:10.3760/cma.j.issn.1001-0939.2020.03.009.

[12] Biot C, Daher W, Chavain N, Fandeur T, Khalife J, Dive D, et al. Design and synthesis of hydroxyferroquine derivatives with antimalarial and antiviral activities. J Med Chem 2006;49:2845–9.

[13] Marmor MF, Kellner U, Lai TY, Melles RB, Mieler WFAmerican Academy of Ophthalmology. Recommendations on Screening for Chloroquine and Hydroxychloroquine Retinopathy (2016 Revision). Ophthalmology Jun 2016;123(6):1386–94 Epub 2016 Mar 16. doi:10.1016/j.ophtha.2016.01.058.

[14] Yao X, Ye F, Zhang M, Cui C, Huang B, Niu P, et al. In Vitro Antiviral Activity and Projection of Optimized Dosing Design of Hydroxychloroquine for the Treatment of Severe Acute Respiratory Syndrom irus 2 (SARS-CoV-2). Clin Infect Dis 2020 Mar 9 pii: ciaa237[Epub doi:10.1093/cid/ciaa237.

[15] Raoult D, Houpikian P, Tissot Dupont H JM, Arditi-D J, Brouqui P. Treatment of Q fever endocarditis: com n of 2 reg s containing doxycycline and ofloxacin or hydr chloro Arch Inte led 1999 Jan 25;159(2):167–73.

[16] Lagier JC, Raoult D. Whipple sease and *Trop pplei* infections: when to suspect them an w to di se and t hem. Curr Opin Infect Dis Dec 2018;31(6): 0 [x] ert consensus on chloroquine phosphate for the treatm t of onavirus monia. Zhonghua Jie He He Hu Xi Za Zhi. Mar 3):185-1 i: 10.3760/cma.j.issn.1001-0939.2020.03.00 :10.1097/QC 000 0000489.

[17] Amrane S, Ti pont H, Douc din C, Hocquart M, Mailhe M, et al. Rap ra nosis and a atory management of suspected COVID-19 c es pres at the infectious diseases referral hospital in Marseille, France, - January o March 1st, 2020: A respiratory virus snapshot. Tr nfect Dis 2020 b ahead of print].

[18] N, Richez M, Raoult D, Chabriere E. Simultaneous UHPLC-UV analysis of hydroxychloroquine, minocycline and doxycycline from serum samples for the therapeutic drug monitoring of Q fever and Whipple's disease. J Chromatogr. B Analyt hnol. Biomed. Life Sci. 2017;1060:166–72.

[19] F, Yu T, Du an G, Liu Y, Liu Z, et al. Clinical course and risk factors for mortality of adult inpatients with COVID-19 in Wuhan, China: a retrospective cohort study. Lancet 2020 Mar 11 pii: S0140-6736(20)30566-3[Epub ahead of print]. doi:10.1016/S0140-6736(20)30566-3.

[20] Tabata T, Petitt M, Puerta-Guardo H, Michlmayr D, Wang C, Fang-Hoover J, et al. Zika virus targets different primary human placental cells, suggesting two routes for vertical transmission. Cell Host Microbe 2016 Jul 13;20(1):155–66 Epub 2016 Jun 30. doi:10.1016/j.chom.2016.06.009.

[21] Madrid PB, Panchal RG, Warren TK, Shurtleff AC, Endsley AN, Green CE, Kolokoltsov A, et al. Evaluation of Ebola Virus Inhibitors for Drug Repurposing. ACS Infect Dis 2015 Jul 10;1(7):317–26 Epub 2015 May 11. doi:10.1021/acsinfecdis.5b00030.

[22] Bosseboeuf E, Aubry M, Nhan T, de Pina JJ, Rolain JM, Raoult D, et al. Azithromycin inhibits the replication of Zika virus. J Antivirals Antiretrovirals 2018;10(1):6–11. doi:10.4172/1948-5964.1000173.

[23] Bacharier LB, Guilbert TW, Mauger DT, Boehmer S, Beigelman A, Fitzpatrick AM, et al. Early administration of azithromycin and prevention of severe lower respiratory tract illnesses in preschool children with a history of such illnesses: A randomized clinical trial. JAMA 2015 Nov 17;314(19):2034–44. doi:10.1001/jama.2015.13896.

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